Supplementary methods

Longitudinal Motor-Imagery EEG Decoding: Frozen Baselines, Supervised Recalibration, and Task-Measurement Reliability

This supplement describes the completed analyses reported in the English working draft, using their frozen protocols, executed source and saved audit records. It adds no experiment, statistical result, adaptation rule or permission to execute. The historical NETBCI material is a separate dataset and analysis record.

S1. Source identity, chronology and eligible windows

The source was the Kumar2024 derivative at NEMAR `nm000177/v1.0.5`, CC BY 4.0, bound by the existing 2,705-file selection manifest. The analysis covered 18 published participant identities, six ordered sessions, 360 EDF recordings and 7,200 event identities. Source size/checksum, local SHA256, annotation/event agreement, channel order, sampling rate, units and frozen numerical window hashes were checked. Complete raw and processed windows had no exact numerical duplicates in the frozen split audit. These checks do not independently establish biological identity or the accuracy of derivative-reported calendar timestamps. Session order and within-session run order were checked against the source acquisition metadata.

Development IDs were sub-1, sub-4, sub-7, sub-10, sub-13 and sub-16. The fixed remaining IDs were sub-2, sub-3, sub-5, sub-6, sub-8, sub-9, sub-11, sub-12, sub-14, sub-15, sub-17 and sub-18. Assignment preceded performance calculations and used every third numeric published ID, starting at one. Later analysis of previously inspected test identities remains exploratory.

Exactly five seconds and 2,560 samples at 512 Hz were required for a complete window. The 44 partial events remained in source/split records and were excluded from tensors without padding, replacement or selecting another run. The included role counts were:

Role	Fixed whole-run assignment	Included windows across 18 IDs
Source training	ses-0, run-0 and run-1	720
Source validation	ses-0, run-2	360
Source query	ses-0, run-3	360
Target support	ses-1 through ses-5, run-0	1,786
Target query	All other target-session runs	3,930

Each person had 40 training and 20 validation windows. The fixed first support run allowed at most 20 events, at most 10 per class, before partial exclusion. Among 90 target sessions, 14 had 19 included support events and 76 had 20. There was no replacement or accumulation of support from other target sessions. The query inventory retained all 4,320 event identities, including 30 partial exclusions, yielding 4,290 included queries. The saved-result comparison verified the same trial IDs, labels, sessions, runs, roles and included-window hashes for every CPU method and every EEGNet seed.

S2. Epoch-local preprocessing and descriptive coordinates

The channel order was F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, C3, Cz, C4, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8 and POz. Labels were left hand = 0 and right hand = 1. MNE numerical

input was in volts; the native physical metadata used microvolts. Each epoch independently underwent:

1. Common-average reference over all 22 channels at each sample.
2. Fourth-order SciPy Butterworth SOS band-pass filtering at 8–30 Hz with `sosfiltfilt` and its default odd padding.
3. `resample_poly(up=1, down=4)` with its default Kaiser-5 window and constant padding, yielding 22 × 640 float32 values at 128 Hz in volts.

These operations fitted no state across epochs. Zero-phase filtering is an offline convention. A raw within-window peak-to-peak value above 250 µV in any channel generated a fixed quality flag. All 1,512 flagged complete windows remained among the 7,156 primary windows. No expert artifact certification or verified reconstruction of the acquisition CPz reference was claimed.

The predefined power description used Welch estimates with a Hann window, 256-sample segments, 128-sample overlap, constant detrending, density scaling, one-sided output and mean averaging. Mu power integrated 8–13 Hz and Beta power 13–30 Hz by trapezoidal integration including band-edge bins. Features were `log10(max(power, float64_tiny))`, ordered channel-major with Mu then Beta, giving 44 features. They are absolute task-window powers in V^2 before the log transform, not baseline-normalized ERD.

Covariance descriptions used a fixed 21 × 22 Helmert basis to remove the CAR null direction. Epochs were time-centered, covariance used the `n−1` denominator, and trace-normalized covariance was shrunk as `0.9*C + 0.1*I/21`. The source-training-only affine-invariant Riemann mean used an arithmetic initializer, unit step, at most 100 iterations and tolerance `1e-10`. Other roles supplied distances to that fixed reference, never a refit. A source-training StandardScaler and full-SVD PCA retained four components without whitening. Primary preprocessing and decoder fits did not change in the fixed quality-flag, count-matched or C3/Cz/C4 descriptive sensitivities.

For the pre-frozen drift associations, the covariance measure was target-query mean distance minus source-query mean distance. The task-axis measure was the one-dimensional Wasserstein distance between target and source query projections onto the fixed source-training right-minus-left power direction. Target labels did not construct that drift variable; labels were used for BA and explicitly supervised retrospective class descriptions. These remaining-cohort associations were descriptive analyses, not development-only reliability probes or policy selection.

S3. Decoder definitions and label budgets

S3.1. CPU baselines

CSP used four components, Ledoit–Wolf regularization, log mean-square features and `norm_trace=False`; LDA used `solver="lsqr"` and automatic shrinkage. The baseline seed was 42 and there was one prespecified configuration. Source-only CSP/LDA fitted source training and then froze both stages. Frozen-CSP/LDA recalibration reused source filters and refitted LDA on source training plus current support. Full refitting fitted both CSP and LDA on that same union. Every fit recorded exact trial IDs; validation and query were forbidden from fitting or selecting the CPU pipeline. Source-query predictions used the same source model for all three method labels.

S3.2. Source-only EEGNet

The official `braindecode.models.EEGNet` implementation used F1=4, D=2, F2=8, temporal kernel 64, depthwise kernel 16, mean-pool sizes 4 and 8, ELU, dropout 0.5, automatic final-convolution length and spatial max norm 1.0. Batch normalization used momentum 0.01, affine parameters and epsilon 0.001. `final_layer_with_constraint=false` and `norm_rate=0.25` were retained exactly as recorded in the executed configuration.

For each participant, channel mean and standard deviation were fitted over source-training trials and time only, with a `1e-8` V scale floor. The optimizer was Adam: learning rate 0.001, betas (0.9, 0.999), epsilon `1e-8`, zero weight decay, no AMSGrad or scheduler. Training

used unweighted cross entropy, no label smoothing, batch size 16, shuffled training rows, `drop_last=false`, `num_workers=0` and at most 100 epochs. The source-validation mean cross entropy selected the checkpoint, with patience 15, minimum improvement zero and the earliest epoch on ties. The complete best state was restored. All seeds 20261012, 20261013 and 20261014 were retained; no favorable seed was selected.

The 54 completed source fits used 2,871 epochs and 8,613 optimizer steps. Eleven reached the 100-epoch cap; reaching that cap was not relabeled as a patience termination. Validation BA/CE was descriptive because validation selected epochs. It received no new population interval or hypothesis test.

For the accepted query evaluation, original package/checkpoint bytes and embedded source context were verified before loading with `torch.load(map_location="cpu", weights_only=True)`. Evaluation mode, disabled gradients and inference mode prohibited optimizer, parameter, batch-normalization and normalization updates. The original state digest was recorded before and after each model's predictions. Labels supported only the fixed metrics; class prediction was untuned logit argmax. The original source CE replay absolute tolerance remained `1e-7`. Independent saved-result audits checked the recorded states and arithmetic without a second model forward.

S3.3. Comparison boundary

Pipeline	Source estimator/optimizer labels	Additional source-validation labels for selection	Current support labels for fitting
Source-only CSP/LDA	40	0	0
Frozen CSP + LDA recalibration	40	0	19/20
CSP/LDA refit	40	0	19/20
Always-frozen EEGNet, each seed	40	20	0

The CPU validation run was not used for model selection or fitting. EEGNet therefore had different source-label use, and only the CPU updates used target support. Equal query identities do not establish equal information or compute budgets. The two CPU update methods shared exact labeled fitting IDs, although updating both stages and updating LDA alone involved different computations. The original full-refit versus frozen contrast deliberately added current support; it was not an equal-target-label architecture comparison.

S4. Endpoints, resampling and evidence hierarchy

BA was the mean of left and right recall. Confusion counts, recalls, predicted class proportions and single-class prediction indicators accompanied each session. CPU target endpoints first averaged five sessions per participant. EEGNet first averaged all three seeds within participant/session, then averaged five target sessions per participant. Source query remained separate. Cohort means weighted participants equally, never pooled trials or treated seeds and sessions as additional independent people. Development6 and remaining12 were separate; all18 was a secondary exploratory combination.

The primary contrast remained full CSP/LDA refit minus source-only CSP/LDA in remaining12, with paired participant resampling. Mean/BA descriptive intervals used 10,000 draws with seed 42 and the 2.5th/97.5th percentiles. The saved-query comparison and decoder endpoint summaries used lexicographic participant-ID order. The six-person reliability bootstrap used numeric order 1,4,7,10,13,16, averaged source folds or support sessions within person first, and

used `default_rng(42)` for 10,000 six-ID samples with replacement. Missing participant values made the entire reliability interval unavailable rather than silently changing its participant denominator.

The original descriptive Spearman drift intervals instead used **2,000 participant-block bootstrap draws**, seed 42. Each selected participant contributed all its session rows together. They did not use 10,000 independent session or trial samples. Both original drift measures retained 2,000 valid block draws. The difference between these two resampling designs is preserved.

No new EEGNet-minus-CPU contrast interval, p value, multiple-testing claim or formal test against 50% was added. The saved comparison reproduced the original CPU primary contrast, +7.42 percentage points [2.49, 12.48], with nine positive and three negative participant differences. The LDA-only contrast remained uncertain. All subsequent same-dataset interpretation remained exploratory after prior test exposure. No task-aware policy benefited from looking at these query outcomes.

S5. Development-only measurement reliability

S5.1. Fixed low-dimensional power probes

The candidates were standardized 44-dimensional Mu/Beta power, its source PCA4 coordinates and six C3/Cz/C4 Mu/Beta coordinates. Only the six development IDs' source training and preassigned current support entered the calculations. Source-run probes trained on run-0's 20 rows and evaluated run-1's 20, then reversed the roles. Each run contained ten events per class. StandardScaler and PCA for a fold were fitted to its TRAIN20 alone. The diagnostic classifier used the right-minus-left class-centroid axis and the midpoint threshold. It did not reuse a transform fitted on the held-out source run.

The exact Cartesian set of 20 × 20 circular label phases was shared across the two source folds and three candidates, including identity. Label-dependent centroid directions and midpoints were recomputed for controls; label-independent fold scales/PCA could be reused. The 14,400 phase-control fold/candidate evaluations included the 36 identity/observed probes; these were auxiliary measurements, not replacements for the primary decoder baseline. All candidates remained; the highest point estimate did not select a winner.

S5.2. Supervised SOURCE20 CSP transfer probes

The completed signed-score run02 fitted CSP and LDA anew on each TRAIN20 fold. Because CSP is label-dependent, every training-label phase refitted both stages. For each participant/fold, 20 training-label phases produced 20 cached models and their held-out predictions. Crossing those predictions with the 20 held-out label phases yielded all 400 joint `(run0_shift, run1_shift)` pairs. Held-out labels never entered fitting, so caching across held-out phases preserved the exact full-pipeline controls.

Across six IDs, this produced 240 fitted pipelines—240 CSP plus 240 LDA fits, including the 12 identity-fold pipelines—4,800 cached held-out trial predictions, 4,800 fold/phase BA evaluations and 2,400 participant/phase-pair twofold means. All twelve source runs had 20 unique circular label assignments. The observed value was the identity phase pair `(0,0)`. All 240 attempted pipelines completed; the implemented expected singular-fit failures would have remained explicit unavailable values, not exclusions of participants or controls.

S5.3. Frozen SOURCE40 coordinates and signed scores

Current support used the six existing SOURCE40 baseline CSP/LDA models without a new decoder fit. For an epoch `X`, the four CSP features were the **natural logarithms** of mean squared projected amplitude:

`f_j(X) = ln(mean_t((W_j X_t)^2)), j=1,...,4.`

The executed consumer rejected nonpositive/nonfinite power rather than inventing features. The vector diagnostic fitted a four-coordinate StandardScaler on SOURCE40 features only, using population variance (`ddof=0`), and applied that fixed transform to support. Its fixed source direction was `d_source = mean(z|right) − mean(z|left)` and its unit direction was `u_source = d_source / ||d_source||` when estimable.

Within each support class, trials were chronologically ordered by acquisition time, start sample and source row. The first `floor(n_class/2)` entered the first half and the remainder entered the second; corresponding class halves were combined. For a half direction `d_h`, the diagnostics retained source/full support cosine, half-to-half cosine and the signed axis contrast

`A_h = (d_h · u_source) / SD_SOURCE40(z · u_source).`

This contrast has **no factor of one half**. Its source projection SD used `ddof=1`. Zero directions, missing classes and non-estimable scales remained explicitly unavailable.

The separate LDA diagnostic used original unstandardized CSP log features, saved coefficient/intercept and classes `[0,1]`:

`s = f @ w + b; prediction = right if s > 0, otherwise left.`

The intercept and zero threshold remained unchanged; support was neither centered nor used to fit a score scale. With `SD_s = SD_SOURCE40(s)` using `ddof=1`, each half retained the balanced true-class signed margin

`M_h = (mean(s|right,h) − mean(s|left,h)) / (2*SD_s).`

The factor of one half is part of this score-margin definition. Separate right margin `mean(s|right,h)/SD_s`, left margin `−mean(s|left,h)/SD_s`, score location, recalls, predicted proportions, BA and collapse were also saved. The vector-axis contrast and LDA margin are distinct measurements. Neither is a calibrated probability or a validated neural-information measure. A positive half-product can arise from two negative margins, and a positive class contrast can coexist with both classes on one side of the frozen threshold.

S5.4. Support controls and denominators

Each current support run supplied all 19 or 20 circular label phases, including identity, with source features, axes, scales and LDA fixed. The primary control recomputed classwise chronological halves after every phase. A separate sensitivity retained the original half indices while rotating labels. For an estimable observation, the descriptive upper-tail fraction was `count(available control >= observed) / count(available control)`, retaining ties and identity without a plus-one adjustment. Attempted, available, unavailable and unique-assignment denominators were recorded explicitly. Unverified stationarity, cyclic boundaries and cue-order dependence prevent treating these fractions as formal permutation p values.

For CSP run02 there were 596 support phases per half method; all 596 were available for each of four saved control metrics—source/full-support cosine, half-to-half cosine, vector half-contrast product and LDA half-margin product— with zero unavailable. For the three power candidates there were 1,788 phases per method, all available for each of three metrics; the two methods contributed 3,576 method/phase evaluations. Phase counts do not add participants. No sign flip, candidate, confidence cutoff or adaptation threshold was selected from the results.

S6. Retained outcomes and historical failures

The neural query analysis retained 16 collapsed seed/session outcomes across seven source-model jobs and seven participants: 14 target and two source outcomes. The frozen CPU model retained three target-session collapses; its two updates retained zero. These denominators differ and their raw counts are not a matched statistical comparison. All participant decreases and all three neural seeds stayed in the aggregates. The five collapsed source-validation fits and eleven 100-epoch terminations also remained in the source records.

CSP run02's observed source twofold mean BA was 62.9167% [52.5, 71.6667]; the participant mean of five support half-cosines was 0.295325 [0.032041, 0.573003]. Among 30 support sessions, signed score half-margins were both positive in 19, both negative in two and opposite in nine; five sessions had frozen-score class collapse. These are calibration-pool measurements, not independent target-query performance of an adaptation rule.

The earlier signed-score run01 actually launched and failed when the resource observer could not read the host's procfs `children` interface. It retained its claim, empty consumer log, termination/exit evidence and private failure backup; there were no saved scientific outputs. An empty log does not prove that no scientific statement or fit began. Explicitly authorized run02 changed process monitoring, preserved the original absolute window and scientific consumer, and completed the 240-pipeline measurement. Its numerical audit reconstructed 629,137 saved-feature arithmetic fields with maximum difference zero; it did not independently recompute EEG projections or refit estimators.

The older, differently specified CSP diagnostic with random-label controls and a mean cosine of 0.298181 is a historical record. It does not replace run02's exact rotations, signed margins, original SOURCE40 reuse and mean 0.295325. The source-fit batch's initial intent-export pause and subsequent same-run resume, and the earlier container replacement, also remain in their original records. Container rebuilding is not described as proof that every unrecorded historical runtime detail was identical. Corrected audit/schema-reader failures remain distinguishable from new scientific attempts. No old receipt or producer stage flag was rewritten to erase these boundaries.

S7. Recorded software, immutable identities and audit coverage

The CPU baseline used Python 3.11.16, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2 and MNE 1.10.2. Its recorded OMP, OpenBLAS and MKL thread limits were one. Source GPU provenance recorded PyTorch/Torchaudio 2.8.0+cu128 and Braindecode 1.2.0; the source runtime inventory and per-fit records retain the observed hardware and container qualifications. The query CPU contract specified the same Python/NumPy/SciPy/sklearn/MNE versions, PyTorch/Torchaudio 2.8.0+cpu and Braindecode 1.2.0, one PyTorch thread, OMP/OpenBLAS limits of one and `CUBLAS_WORKSPACE_CONFIG=:4096:8`. GPU fitting and CPU inference are separate stages; no performance or compute-budget equivalence is inferred.

Artifact	Full-byte SHA256
CPU frozen protocol	`ac290f0e2c5ca58b3745d7cccd83c3fd265e5be9978f461ea5ed0fe7f645831e`
CPU scientific configuration	`f1b2d0fa870493cc2aaab906fe6d8b1227e45b2a270e06700e3ca137fbfe9c89`
Frozen split	`ac6e9f2cdd66b3c26814cfc04edf5f8e08eb592ca523c7ab029a2fa56ffd0b28`
Executed EEGNet configuration	`14c291ca9762cb676a5f61608ab7b895435fdb957b7396056093c8310435856d`
Frozen query protocol	`061892404460a97dd1d999cec3067a0947b2eb7fec063293cb2e65ee5bf5729c`
Query operation contract	`fb3a429776e77a4ee8de96c8c7f2a833a4d311ced d8f0a9bcec3d0c6e8df225a`
Executed signed-score scientific consumer	`ebc866734add9769c7b881332cb9362badaa0f6bf401bc2c6dfa717b645960f3`
Accepted source54 ledger	`013668c8d1b1c5d6061f3701d495cf7770743a1c6fda8678f5db7ffd50992670`
Original query full-GET receipt	`15682a07c13d69aa7bd7aca2cf65327aea7c58fb4e8cb1205b195d4b0fcbb9e2`
Saved-query comparison receipt	`a45664d75a3b1e84373f4af809715a8adb356a7710ba14d0843894e968cd80ae`

Historical `execution_enabled=false` and other preparation flags in a frozen contract were not silently edited. Actual execution authorization, launcher receipts, audit, private R2 verification and parent acceptance were recorded separately. The query ran once at commit `d34703ca5bccf7a5b88a91c2ffb2cba893f7cec8`; its audit delivery was fixed at `ac5a1f6d014421f0adf6f924a58a7063558b89f7`. The original operator verified 147 private evidence/result objects, 17,608,200 bytes, by full GET/SHA256 with zero failures. Actual parent acceptance at 2026-10-15T09:49:05Z checked the complete receipt, identities and saved arithmetic; it did not claim a new parent network GET, a second forward or rehashing all 147 original objects.

Evidence layer	Existing record and practical boundary
Original CPU results and fit access	Execution receipt; exact source/support IDs and unchanged primary contrast
Source54 saved validation	Aggregation receipt; no new fit, forward or query from this aggregation
Low-dimensional reliability	Executed report, frozen plan; development source/support only
Complete supervised CSP reliability	Run02 report, scientific summary, methods review; saved-feature replay rather than a second EEG projection
Query identity and arithmetic	Query audit report; all54 original archives and 1,189 inputs bound; independent numerical reconstruction without forward
Query acceptance	Actual parent ACK; source54 ledger and historical producer flags unchanged
Matched saved-query aggregation	Comparison report and exact cohort table; no newly fitted or inferred strategy

The original source EEG, checkpoint tensors and binary measurement artifacts remain outside GitHub. Public source access and data terms are distinct from the repository's software license and institutional secondary-data requirements. The author review checklist lists the outstanding author-supplied publication details. Completing that checklist does not change these methods or establish a task-aware, human-learning, retention or clinical effect.